

Tab 1

Pre-requisites

1. FastAPI Framework Knowledge: FastAPI Documentation
2. Google Gemini API Familiarity: Google Generative AI Documentation
3. Streamlit for Frontend: Streamlit Documentation
4. Python Programming Proficiency: Python Official Docs
5. Environment Setup: `pip install fastapi uvicorn streamlit python-docx fpdf Pillow requests google-generativeai python-dotenv`

Project Workflow

Activity 1: Model Selection and System Architecture

1.1: Research and select the Model: Gemini 1.5 Pro model from Google Generative AI for legal document generation.

1.2: Define architecture:

- o Frontend: Streamlit for user input and document download.
- o Backend: FastAPI (`main.py`, `routes.py`) to handle API requests.
- o AI Integration: Gemini 1.5 API used inside `ai_core/gemini_generator.py`.

1.3: Set up development environment with `.env` file containing the Gemini API key.

Activity 2: **Core Functionalities Development**

2.1: Implement core generation features:

- o Legal document creation
- o Editable text preview
- o Format & export as `.txt`, `.docx`, and `.pdf`

2.2: Build FastAPI backend:

- o Endpoint `/generate` to process POST requests
- o Accepts `document_type`, `parties`, `terms`, `dates`
- o Returns generated text from Gemini